Automated Truck Parking Spatial Occupancy Detection Using 3D Perception

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A scalable truck parking occupancy detection system utilizing monocular depth estimation with VGGT. Evaluated across 1,154 samples from 16 public truck parking sites, the pipeline achieves 94.5% accuracy on undisciplined lots and 95.8% on disciplined lots, providing a lower-sensing alternative to multi-camera SfM reconstruction with comparable accuracy.

Inputs & Outputs:

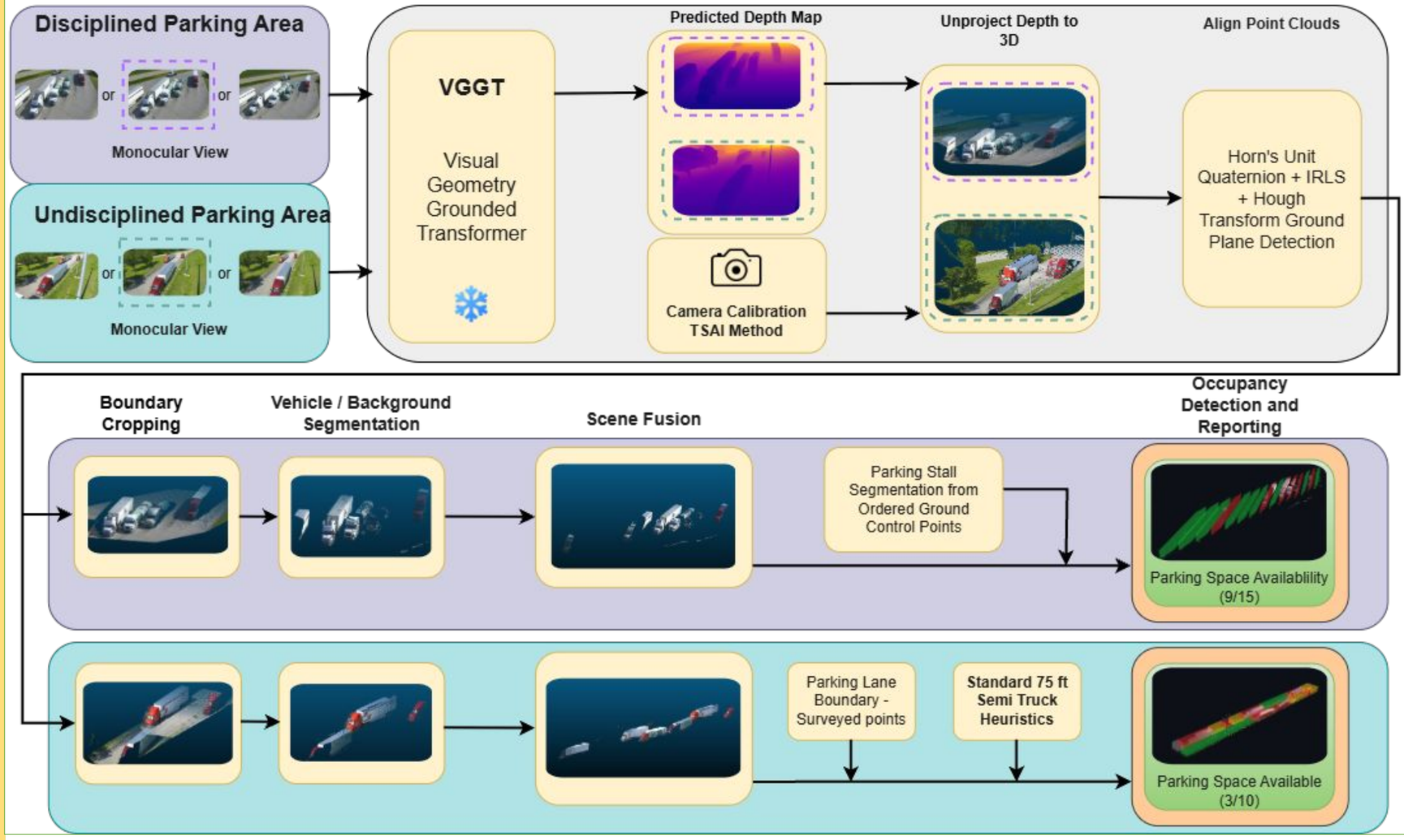
- **One-Time Inputs:** Lot/lane boundaries, surveyed ground control points, camera calibration parameters
- **Recurrent Inputs:** Hourly overlapping camera samples per site
- **Internal Outputs:** Predicted monocular depths, 3D point cloud reconstructions
- **User Outputs:** Occupancy information

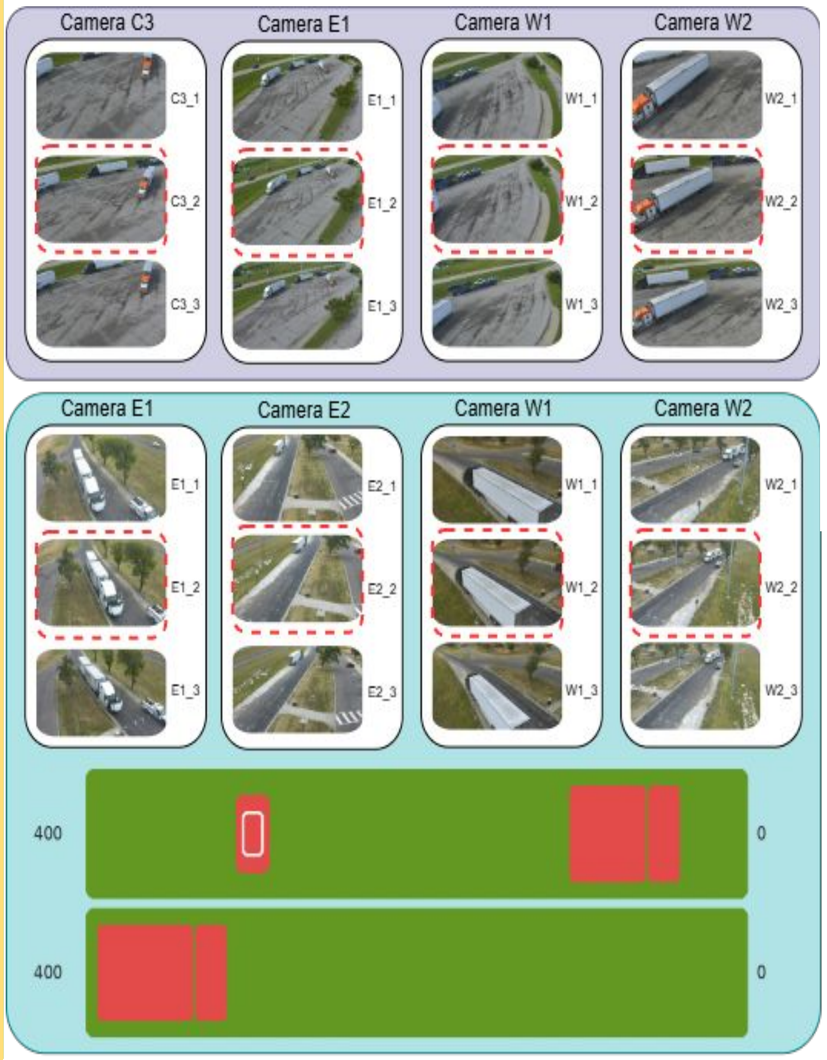
Truck Parking Occupancy Estimation Pipeline:

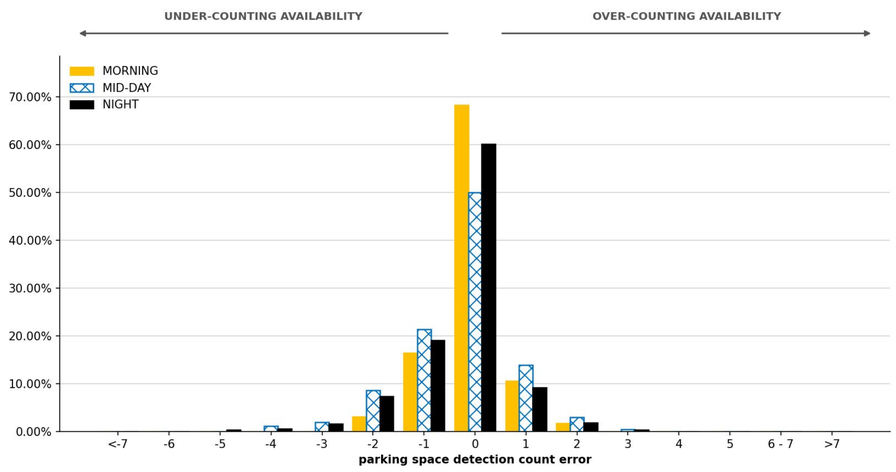
- **Depth estimation:** Estimate depth maps from images using VGGT.
- **Camera calibration:** Create camera intrinsics and extrinsics using the Tsai method.
- **Point cloud reconstruction:** Unproject 3D point clouds from depth maps.
- **Statistical outlier removal:** Apply removal using 20 nearest neighbors and a 2.0 standard-deviation ratio.
- **Survey registration:** Register point clouds to surveyed ground control points.
- **Spatial filtering:** Remove points outside the parking lot using an empirical boundary.
- **Ground and vegetation removal:** Remove vegetation by color, and ground/floating fragments via height-banded plane search.
- **Lane-level occupancy classification:** Estimate available parking using point cloud density and planar boundary extents.

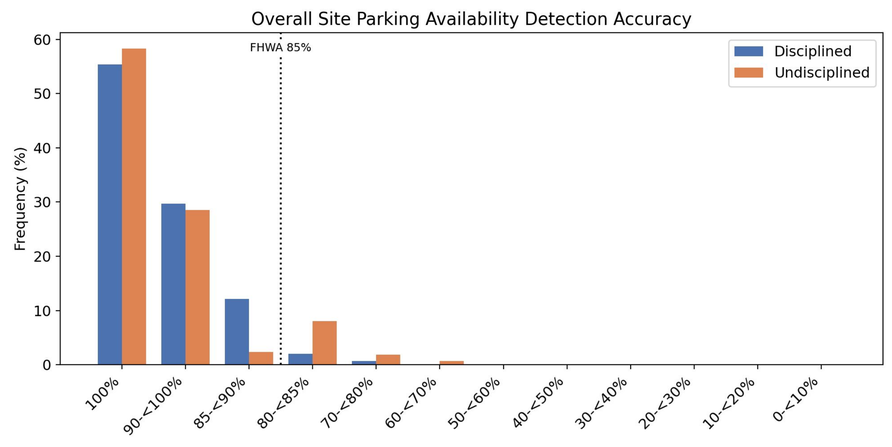
Dataset / Experiment:

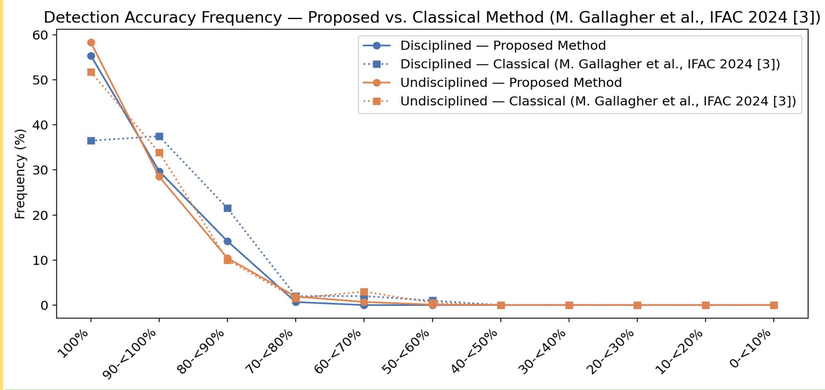
- **Sites and Data:** Evaluation dataset contains over 1,000 ground truth labeled observation samples collected from 16 different sites evenly spaced across a 24-hour period.
- **Camera Configurations:** Current configuration uses three PTZ cameras per pole; the proposed pipeline uses one camera.
- **Ground Truth Generation:** Disciplined parking lots were marked "Occupied" or "Empty"; in undisciplined parking lots, occupied space lengths and spatial separations were recorded.
- **Evaluation:** Accuracy is measured against manually labeled ground truth data.

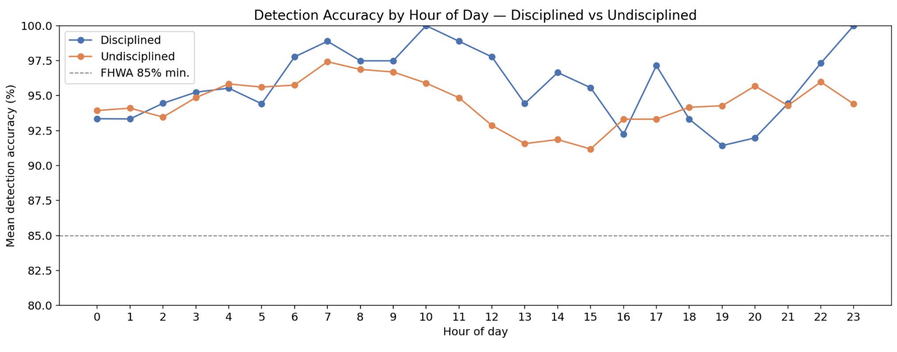




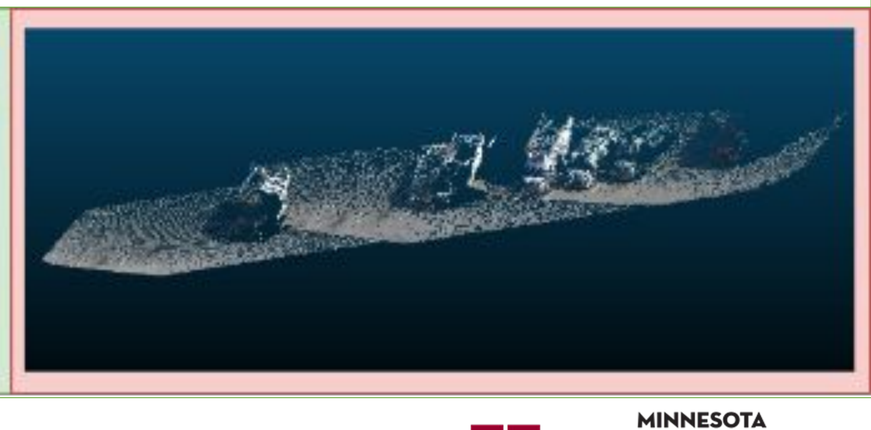












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